Exploring Patterns of Environmental Justice

EDS 223 Homework 2

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```
# Library of Packages
library(tidyverse)
library(sf)
library(here)
library(stars)
library(tmap)
library(quarto)
library(gt)
library(cowplot)
```

```
# Set Theme
theme_set(theme_bw())
```

Part 1: Legacy of redlining in current environmental (in)justice

Your first task is to explore historical redlining in Los Angeles and its legacy on present-day environmental justice.

Description

For this assignment, you must complete the following:

```
# Read in geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(here::here(
  "data",
  "ejscreen",
  "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"
))
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `C:\meds\eds223\eds223-hw2\data\ejscreen\EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```r
# Filter to Los Angeles County, CA
la_co <- ejscreen |>
  dplyr::filter(CNTY_NAME %in% c("Los Angeles County"))

# Read in mapping inequality for Los Angeles
mi_la <- sf::st_read(here::here(
  "data",
  "mapping-inequality",
  "mapping-inequality-los-angeles.json"
))
```

```
Reading layer `mapping-inequality-los-angeles' from data source
  `C:\meds\eds223\eds223-hw2\data\mapping-inequality\mapping-inequality-los-
angeles.json'
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

```r
# Fixing HOLC geometries
holc <- st_make_valid(mi_la)

# Transfrorming mapping inequality CRS
mi_la_transform <- st_transform(mi_la, crs = st_crs(la_co))

# Checking for matching CRS
if (st_crs(la_co) == st_crs(mi_la_transform)) {
  print('Match')
} else {
  warning('NOT A MATCH')
}
```

```
[1] "Match"
```

```
# Joining mapping inequality and LA EJScreen
joined_la <- st_join(la_co, mi_la_transform)
```

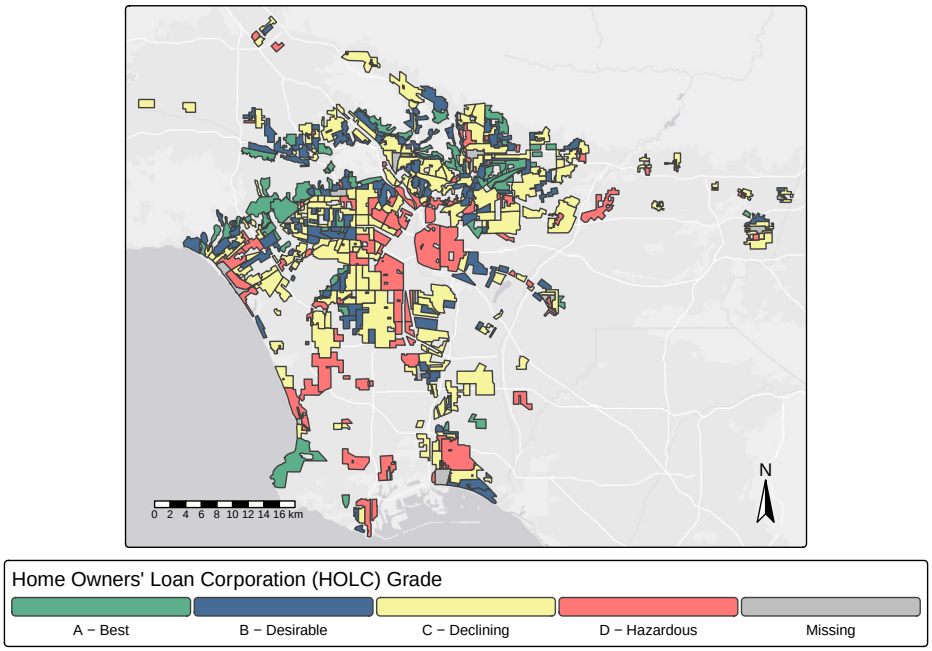
1. Create a map of historical redlining neighborhoods, including:

- neighborhoods colored by HOLC grade
- an appropriate base map
- a clear title and legend

```
# Setting colors for HOLC grade levels and text for HOLC grade labels
grade_colours <- c('#5DAE8B', '#466C95', '#F6F49D', '#FF7676', '#9b9090')
grade_labels <- c('A - Best', 'B - Desirable', 'C - Declining', 'D - Hazardous')
```

```
# Mapping historically redlined neighbourhoods (HOLC Grade by Census Block Group)
tm_shape(mi_la) + # HOLC polygons
  tm_polygons(
    fill = 'grade',
    fill.scale = tm_scale_categorical(
      labels = grade_labels,
      values = grade_colours
    ),
    fill.legend = tm_legend(
      title = "Home Owners' Loan Corporation (HOLC) Grade",
      orientation = 'landscape',
      width = 50
    )
  ) +
  tm_scalebar(position = c('left', 'bottom')) +
  tm_compass() +
  tm_title(
    "Home Owners' Loan Corporation (HOLC) Grades by Census Block Group \n Los Angeles County
    size = 1.25,
    fontface = 'bold'
  ) +
  tm_layout(component.autoscale = FALSE) +
  tm_shape(la_co) + # LA County basemap
  tm_basemap('Esri.WorldGrayCanvas')
```

Home Owners' Loan Corporation (HOLC) Grades by Census Block Group
Los Angeles County, California, USA



2. Create a table summarizing:

- the percentage of census block groups that fall within each HOLC grade
- Also include the percent of census black groups that don't fall within a HOLC grade in your table
- Hint: The HOLC data contains the grades and the EJScreen data contains the census blocks, so you will need to combine the data spatially before doing summary statistics. Once you combine and no longer need the geometries, you can use st_drop_geometry().

```
# Creating dataframe for percent of block groups in each HOLC grade
percent_block_grade <- joined_la |>
  st_drop_geometry() |>
  count(grade) |>
  mutate(pct_grade = round(n / sum(n) * 100, 2)) |>
  select(-n) |>
  rename(Grade = grade, Percent = pct_grade)

# Displaying dataframe as a table
percent_block_grade |>
  gt() |>
```

Percent Census Blocks with each HOLC Grade

Grade	Percent
A	5.00
B	13.79
C	34.02
D	14.98
NA	32.22

```
  tab_header(
    title = 'Percent Census Blocks with each HOLC Grade'
  )
```

3. Create at least two visualizations summarizing current conditions (from the EJScreen data) within HOLC grades using the mean of the following variables (you may combine variables in a single visualization or look at a single variable per plot):

- % low income
- percentile for Particulate Matter 2.5
- percentile for low life expectancy
- Use ggplot for your visualizations! You will first need to calculate mean of each variable grouped by HOLC grade.

```
# Creating summary dataframe containing means of each current condition variable by grade
summary_la <- joined_la |>
  group_by(grade) |>
  drop_na(LOWINCPCT, P_PM25, P_LIFEEXPPCT) |>
  summarise(
    avg_lowinc = mean(LOWINCPCT),
    avg_pm25 = mean(P_PM25),
    avg_lifexpv = mean(P_LIFEEXPPCT)
  )
```

```
# Plotting summary of means of each current condition variable by grade

# Create fake plot for legend
mock_plot <- summary_la |>
  ggplot(aes(x = grade, y = avg_lifexpv, fill = grade)) +
  geom_col() +
  scale_color_manual(
```

```
    name = 'HOLC Grade',
    values = grade_colours,
    labels = grade_labels
  ) +
  scale_fill_manual(
    name = 'HOLC Grade',
    values = grade_colours,
    labels = grade_labels
  ) +
  theme(
    legend.title = element_text(size = 12, face = 'bold'),
    legend.text = element_text(size = 10),
    legend.key.size = unit(0.5, 'cm'),
    legend.direction = 'horizontal'
  )

# Create joint legend using fake plot
legend <- get_legend(mock_plot)

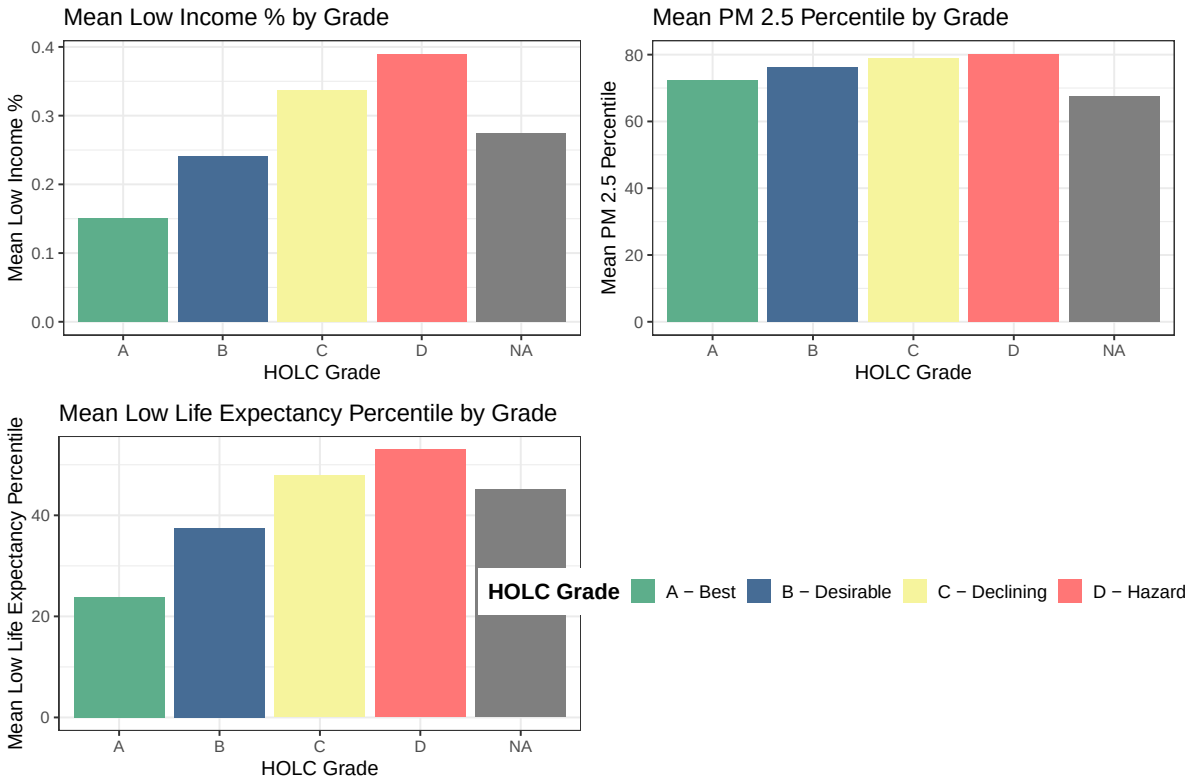
# Low income plot
inc <- summary_la |>
  ggplot(aes(x = grade, y = avg_lowinc, fill = grade)) +
  geom_col() +
  labs(
    title = 'Mean Low Income % by Grade',
    x = 'HOLC Grade',
    y = 'Mean Low Income %'
  ) +
  scale_fill_manual(values = grade_colours) +
  theme(legend.position = 'none')

# PM 2.5 plot
pm <- summary_la |>
  ggplot(aes(x = grade, y = avg_pm25, fill = grade)) +
  geom_col() +
  labs(
    title = 'Mean PM 2.5 Percentile by Grade',
    x = 'HOLC Grade',
    y = 'Mean PM 2.5 Percentile'
  ) +
  scale_fill_manual(values = grade_colours) +
  theme(legend.position = 'none')
```

```r
# Left expectancy plot
lex <- summary_la |>
  ggplot(aes(x = grade, y = avg_lifexpv, fill = grade)) +
  geom_col() +
  labs(
    title = 'Mean Low Life Expectancy Percentile by Grade',
    x = 'HOLC Grade',
    y = 'Mean Low Life Expectancy Percentile'
  ) +
  scale_fill_manual(values = grade_colours) +
  theme(legend.position = 'none')

# Joining 3 plots and legend
current_conditions_summary <- plot_grid(
  inc,
  pm,
  lex,
  legend
)

# Display joint plot
current_conditions_summary
```



4. Write a brief paragraph reflecting on these results

- Interpret the patterns you observe in your results
- Discuss potential relationships between historical redlining grades and current environmental/socioeconomic conditions

With each variable describing current conditions, there is a rise in mean values from HOLC grades A to D, where grade D has the highest mean low income percentage, PM 2.5 percentile, and low life expectance percentile, and grade A has the lowest. The means for low income percentage and low life expectance percentile are larger for census blocks in LA County without a grade than those in grades A and B, however the mean for PM 2.5 percentile is lowest in census blocks with now grade designation. The mean PM 2.5 percentile is likely the lowest in ungraded census blocks, as the majority of graded blocks are located within the urbal sprawl of Los Angeles, whereas there are large numbers of ungraded blocks in less-populated, less dense areas of the county. As a census block is graded more poorly and redlined, the higher the census blocks current condition marker variables.

Part 2: Legacy of redlining in pedstrian and bike accidents Your second task is to explore the legacy of historical redlining in Los Angeles on the report of pedestrian and bike crashes/

fatalities.

Description

For this assignment, you must produce the following based on observations from 2020-2025:

1. A figure summarizing the percent of observations within redlined neighborhoods within each HOLC grade

Create a visualizations that shows:

- The percentage of bike and pedestrian crashes and fatalities within each HOLC grade
- Include an appropiate title, axis labels, and legend
- Hints: Make sure to filter for bike and pedestrian crahes(PEDESTRIAN_ACCIDENT and BICYCLE_ACCIDENT). Look at the difference in NAs between POINT_X/POINT_Y and LONGITUDE/LATITUDE before you decide which to use. Ensure the SWITRS data and HOLC dataset have matching CRS', then perform a spatial join to assign each crash to a corresponding HOLC grade.

```
crashes <- read_csv(here::here("data", "SWITRS_LA", "Crashes.csv")) |>
  as_tibble() |>
  drop_na(POINT_X, POINT_Y) |>
  rename(long = POINT_X, lat = POINT_Y) |>
  st_as_sf(coords = c("long", "lat"), crs = 4326, remove = FALSE)
```

::: {.cell}

```
# Check that crs match before joining
if (st_crs(holc) == st_crs(crashes)) {
 print('Match')
} else {
 warning('NOT A MATCH')
}
```

::: {.cell-output .cell-output-stdout}

```
[1] "Match"
```

::: :::

::: {.cell}

Accident Counts by HOLC Grade and Percentages

grade	count	percent
A	360	2.048947
B	1782	10.142288
C	9026	51.371656
D	4532	25.793967
NA	1870	10.643142

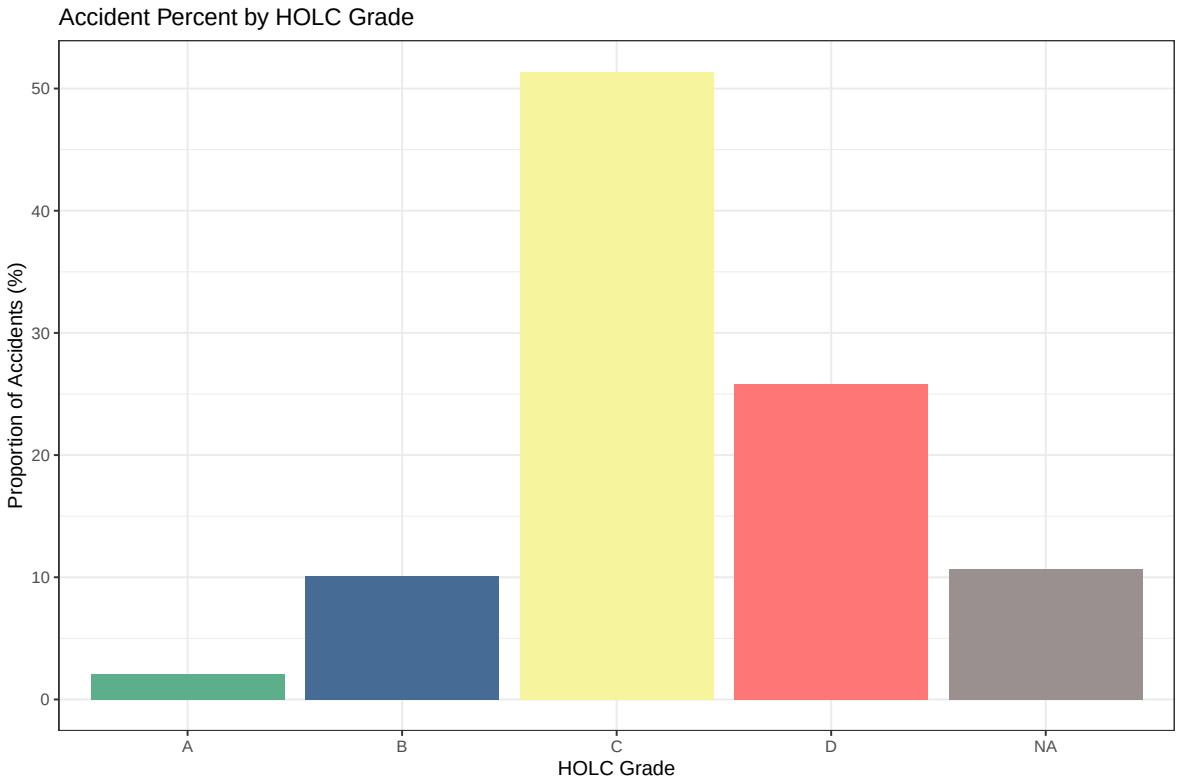
```
# Join crashes and holc, drop geometry, filter for bicycle and pedestrian accidents
crashes_holc <- st_join(holc, crashes, join = st_intersects) |> st_drop_geometry() |> filter
```

:::

```
# Create summary dataframe of accident counts by grade and respective percentages
crashes_holc_summary <- crashes_holc |>
  group_by(grade) |>
  summarise(count = n()) |>
  mutate(percent = count / sum(count) * 100)

# Display summary df as table
gt(crashes_holc_summary) |>
  tab_header(
    title = 'Accident Counts by HOLC Grade and Percentages'
  )
```

```
# Plotting crash summary
ggplot(crashes_holc_summary, aes(x = grade, y = percent)) +
  geom_col(fill = grade_colours) +
  labs(
    title = 'Accident Percent by HOLC Grade',
    x = 'HOLC Grade',
    y = 'Proportion of Accidents (%)'
  )
```


Accident Percent by HOLC Grade
Proportion of Accidents (%)
0
10
20
30
40
50
A
B
C
D
NA
HOLC Grade